

ISA 444/544: Business Forecasting

15: Stationarity, Differencing and Unit Root Tests

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 Automated Scheduler for Office Hours

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Learning Objectives for Today's Class

- Define a **stationary time series**.
- Understand how plotting techniques can help identify non-stationarity.
- Understand how we can **stabilize a non-stationary time series to achieve stationarity** (differencing to stabilize the mean, and log transformation to stabilize the variance).
- Understand how to use **unit root tests** to test for stationarity.

What is Stationarity?

Stationarity

Definition: If $\{y_t\}$ is a **stationary time series**, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

- This means that a **stationary time series**:
 - is roughly horizontal (i.e., no trend)
 - has a constant variance (i.e., no heteroskedasticity)
 - no patterns are predictable in the long-term (i.e., no seasonality)

Class Activity: Stationarity or Non-Stationarity?

Task

KR

Δ KR

Strikes

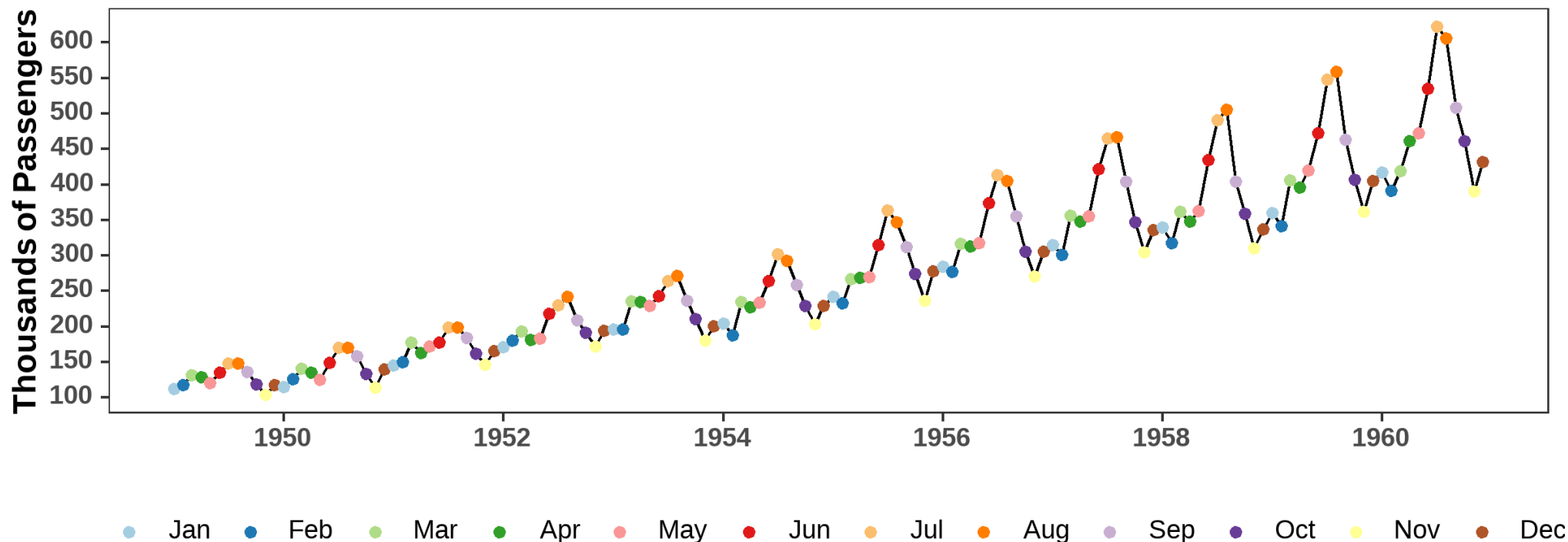
House Sales

Lynx

Beer

Travel

Monthly U.S. Airline Passengers (1949-1960)



How to Identify Non-Stationarity in the Mean

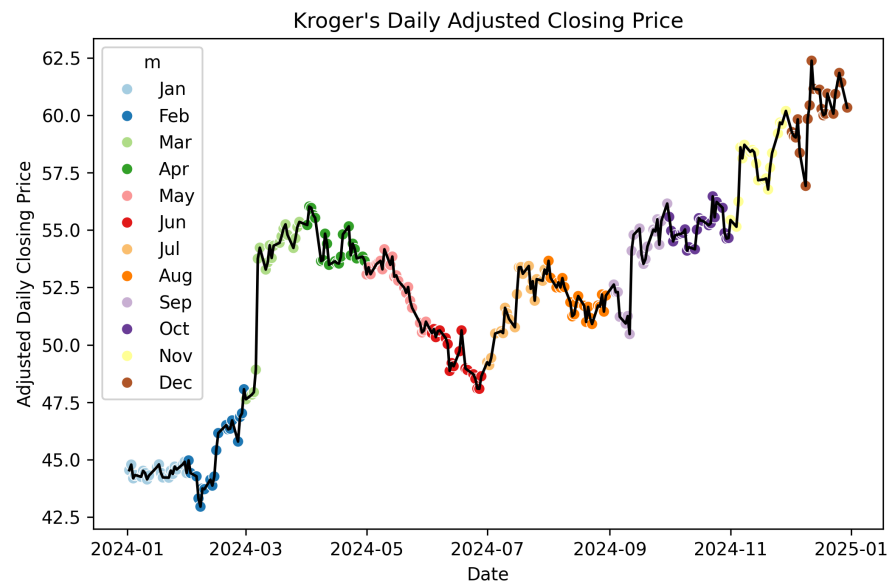
- **Visual Inspection** of the **time series plot (line-chart)**:
 - Look for trends in the time series plot.
 - Look for seasonality in the time series plot.
- Examine the **ACF** of the time series:
 - A **slow decay** in the ACF plot → **non-stationary**.
 - On the other hand, if the ACF plot **drops to zero quickly** → **likely stationary**.
 - **Note:** For non-stationary time series, the **value of r_1 is often large and positive**.

Example: The Non-Stationarity in Kroger Stock Price

```
import pandas as pd
import datetime as dt
import seaborn as sns

path = "../../../data/kroger_stock_2024.csv"
kr = (
    pd.read_csv(path)
    .assign(
        ds=lambda x: pd.to_datetime(x['date']),
        m=lambda x: x['ds'].dt.month_name().str[:3],
    )
)

plt.figure(figsize=(8, 5))
ax = sns.lineplot(
    data=kr, x='ds', y='price', color='black'
)
sns.scatterplot(
    data = kr, x = 'ds', y = 'price', hue = 'm',
    s=40, ax=ax, palette='Paired'
)
ax.set(
    title="Kroger's Daily Adjusted Closing Price",
    xlabel="Date",
    ylabel="Adjusted Daily Closing Price",
)
```



Example: The Non-Stationarity in Kroger Stock Price

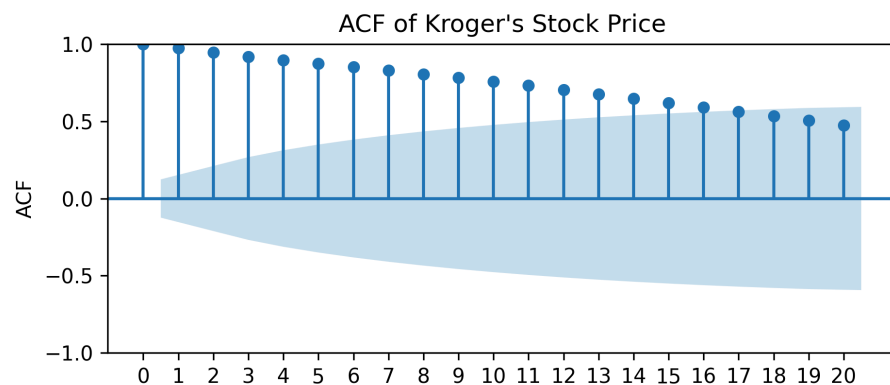
```
from statsmodels.tsa.stattools import acf
from statsmodels.graphics.tsaplots import plot_acf

# Using the data from Slide 7
acf_df = pd.DataFrame(
    {'lag': range(0, 21),
     'acf': acf(kr['price'], nlags=20)}
)

# Print the first few rows of the ACF df
acf_df.head()

# Plot the ACF and beautify the plot
plt.figure(figsize=(8, 5))
plot_acf(kr['price'], lags=20)
plt.xticks(ticks=range(0, 21))
plt.xlabel('Lag')
plt.ylabel('ACF')
plt.title("ACF of Kroger's Stock Price")
plt.show()
```

```
##      lag      acf
## 0      0  1.0000
## 1      1  0.9737
## 2      2  0.9467
## 3      3  0.9189
## 4      4  0.8954
```



How Does the ACF of a Stationary Time Series Look?

- For a **stationary time series**, the **ACF** plot will **decay (cut off) quickly** to zero.
- Non-significant autocorrelations are within the **blue shaded region**, and are **statistically not different from zero**.

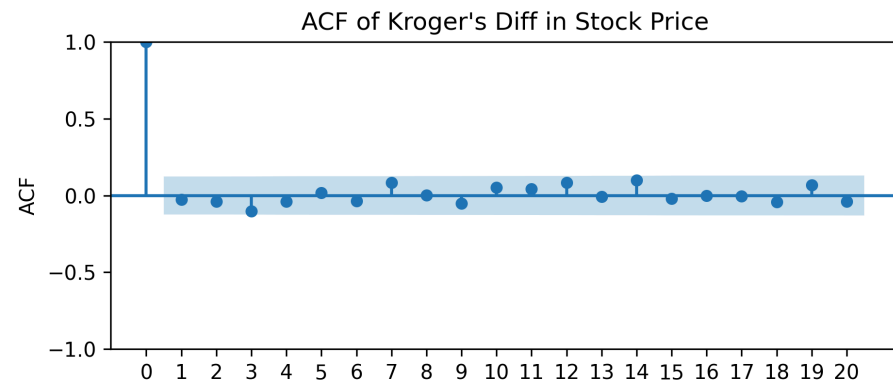
```
kr['delta'] = kr['price'].diff()

acf_df = pd.DataFrame(
    {'lag': range(0, 21),
     'acf': acf(
         kr['delta'], nlags=20, missing='drop'
     )}.round(4)

acf_df.head()

plt.figure( figsize=(8, 5) )
plot_acf(kr['delta'], lags=20, missing='drop')
plt.xticks( ticks=range(0, 21) )
plt.xlabel('Lag')
plt.ylabel('ACF')
plt.title("ACF of Kroger's Diff in Stock Price")
```

##	lag	acf
## 0	0	1.0000
## 1	1	-0.0274
## 2	2	-0.0382
## 3	3	-0.1016
## 4	4	-0.0387



Mathematical Background on the 95% Limits of the ACF Plot

(Understanding how `statsmodels`
computes confidence intervals)

1. Standard Error of ACF for White Noise

For an **uncorrelated white noise** process, the standard error (SE) of the sample ACF at lag (k) is:

$$\text{SE}(\hat{\rho}_k) \approx \frac{1}{\sqrt{N}}$$

- Here, (N) is the **sample size**.
- Assumes the series is **white noise**: no autocorrelation beyond lag 0.
- The **95% confidence interval (CI)** is:

$$\pm 1.96 \times \underbrace{\frac{1}{\sqrt{N}}}_{\text{Standard Error (SE)}}$$

💡 **Key idea:** More data $\longrightarrow$ **Smaller SE** $\longrightarrow$ **Narrower confidence bands**.

2. Bartlett's Formula for SE of ACF

For a **stationary** time series, the **SE must account for autocorrelation**:

$$\text{SE}(\hat{\rho}_k) \approx \sqrt{\frac{1}{N} \left(1 + 2 \sum_{j=1}^{k-1} \hat{\rho}_j^2 \right)}$$

The **sum of squared autocorrelations**, $\underbrace{2 \sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{Effect of past lags}}$, increases SE, which leads to **wider**

confidence bands.

Hence:

- If the time series is **highly correlated** (e.g., a **random walk**), SE **increases**.
- **Differencing reduces autocorrelation**, making the sum **smaller** and the SE **lower**.

3. Why Do CIs Widen at Larger Lags?

Even though **autocorrelations shrink at higher lags**, the confidence intervals still widen. **Why?**

$$\text{SE}(\hat{\rho}_k) \approx \sqrt{\frac{1}{N} \left(1 + 2 \sum_{j=1}^{k-1} \hat{\rho}_j^2 \right)}$$

Larger lags $\longrightarrow$ $\left(\underbrace{1 + 2 \sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{Cumulative sum increases}} \right) \longrightarrow$ **Higher SE.**

Fewer overlapping observations at higher lags $\longrightarrow$ **Increased uncertainty.**

Result: Confidence intervals **widen at larger lags**, even if $\hat{\rho}_k$ is small.

4. Impact of Differencing on Confidence Limits

When **differencing** a time series:

$$Y_t = X_t - X_{t-1}$$

- **Reduces autocorrelation**, shrinking the **Bartlett sum** in SE computation:

$$\underbrace{\sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{smaller after differencing}}$$

- **Decreases SE**, leading to **narrower CIs** compared to the original series.
- Explains **why** `plot_acf()` **shows**:
 - **Wider limits for a trending or non-stationary series.**
 - **Narrower limits after differencing.**

Key Takeaways

- ✓ Original series with **strong autocorrelation** → **Wider confidence bands**
- ✓ Differenced series with **weaker autocorrelation** → **Narrower confidence bands**
- ✓ **CIs widen at larger lags** due to **accumulating estimation uncertainty**

Stabilizing Non-Stationary Time Series

Stabilizing the Mean: Differencing

- **Differencing** is a common technique to **stabilize the mean** of a time series.
- The **first difference** of a time series is the **change** between consecutive observations in the original series:

$$\bullet \Delta y_t = \underbrace{y'_t}_{\substack{\text{First} \\ \text{Difference}}} = \underbrace{y_t - y_{t-1}}_{\substack{\text{Current value} \\ \text{minus previous value}}}$$

- The **differenced series** will have only $N - 1$ observations since it is not possible to calculate a difference y'_1 for the first observation.

Second-order Differencing

Occasionally the differenced data will not appear stationary and it may be necessary to difference the data a second time:

$$\begin{aligned} \underbrace{y_t''}_{\text{Second Difference}} &= \underbrace{y_t' - y_{t-1}'}_{\text{First Difference of the First Difference}} \\ &= \underbrace{(y_t - y_{t-1})}_{\text{First Difference at } t} - \underbrace{(y_{t-1} - y_{t-2})}_{\text{First Difference at } t-1} \\ &= \underbrace{y_t - 2y_{t-1} + y_{t-2}}_{\text{Final expanded form}} \end{aligned}$$

- y_t'' will have $T - 2$ values.
- In practice, it is unlikely to go beyond second-order differences (a different representation of the **data generation process** may be needed).

Seasonal Differencing

A **seasonal difference** is the difference between an observation and the corresponding observation from the previous year.

$$y'_t = y_t - y_{t-m}$$

where m = number of seasons.

- For monthly data $m = 12$.
- For quarterly data $m = 4$.
- Seasonally differenced series will have $(T-m)$ observations.

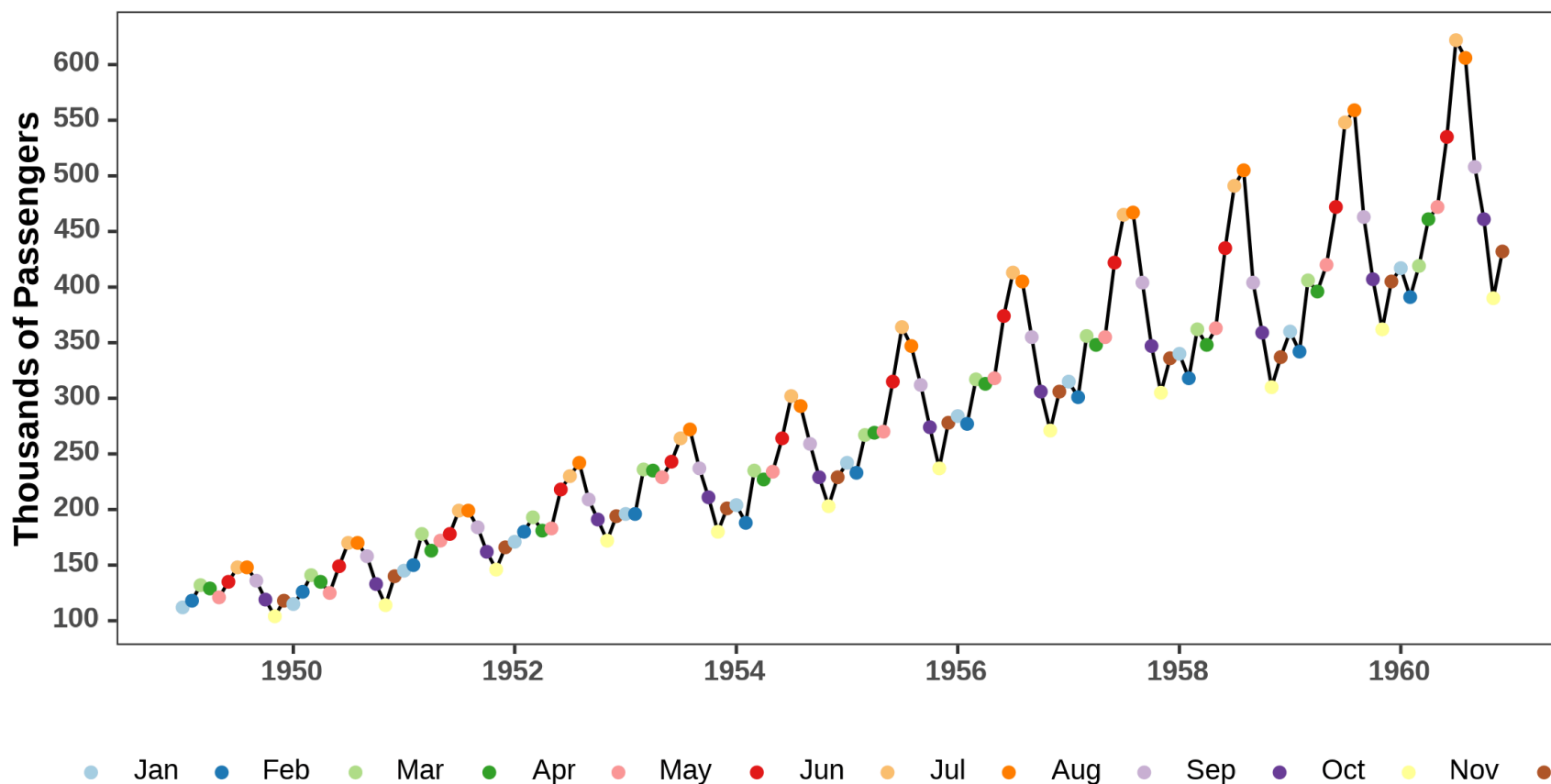
Interpretation of Differencing

- **First differences** are the change between **one observation and the next**.
- **Seasonal differences** are the change between **one observation and the same observation in the previous year** (i.e., year over year changes or changes between one year to the next).

Stabilizing the Variance: Log Transformation

Log transformation is a common technique to **stabilize the variance** of a time series.

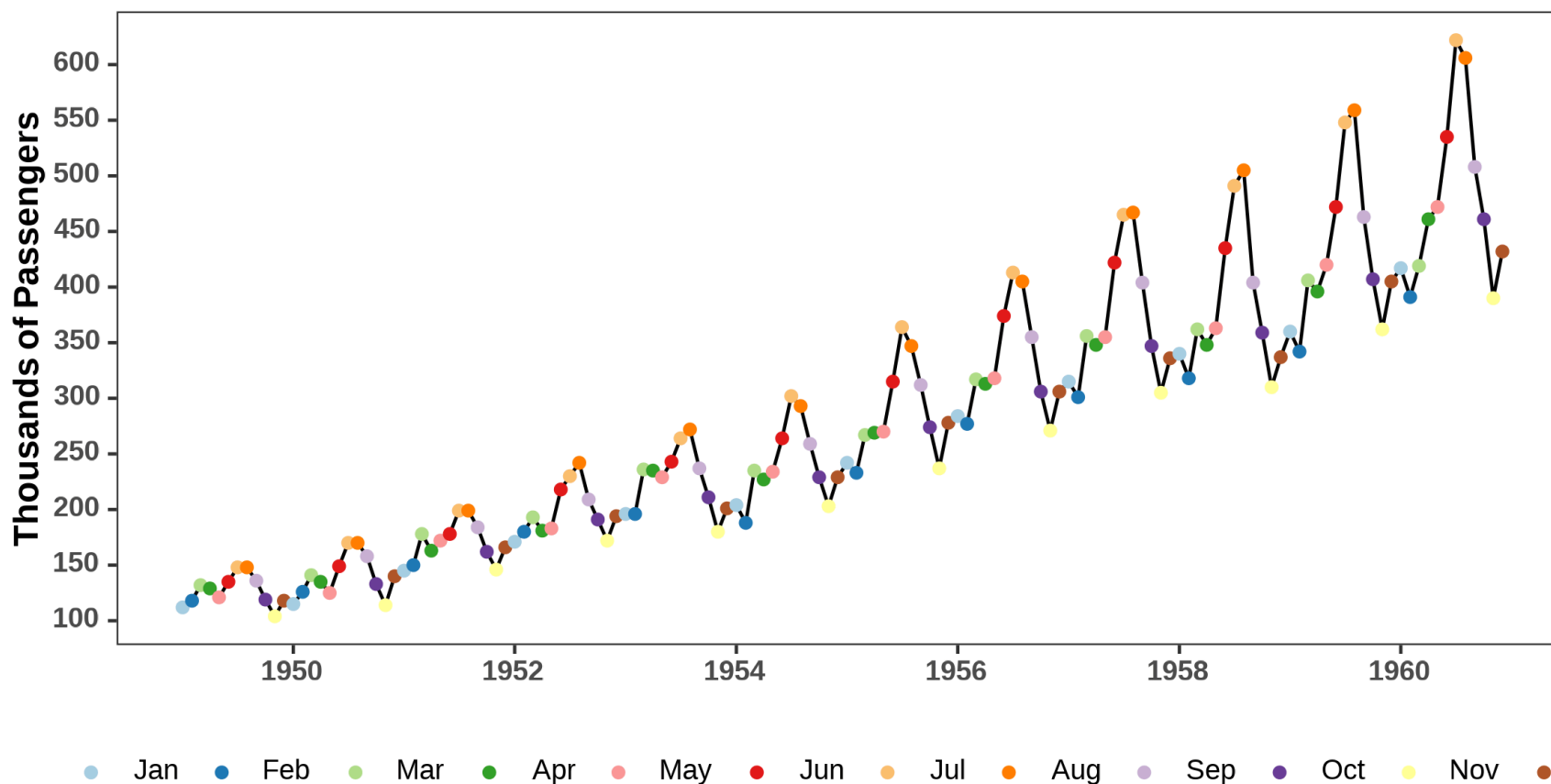
Monthly U.S. Airline Passengers (1949-1960)



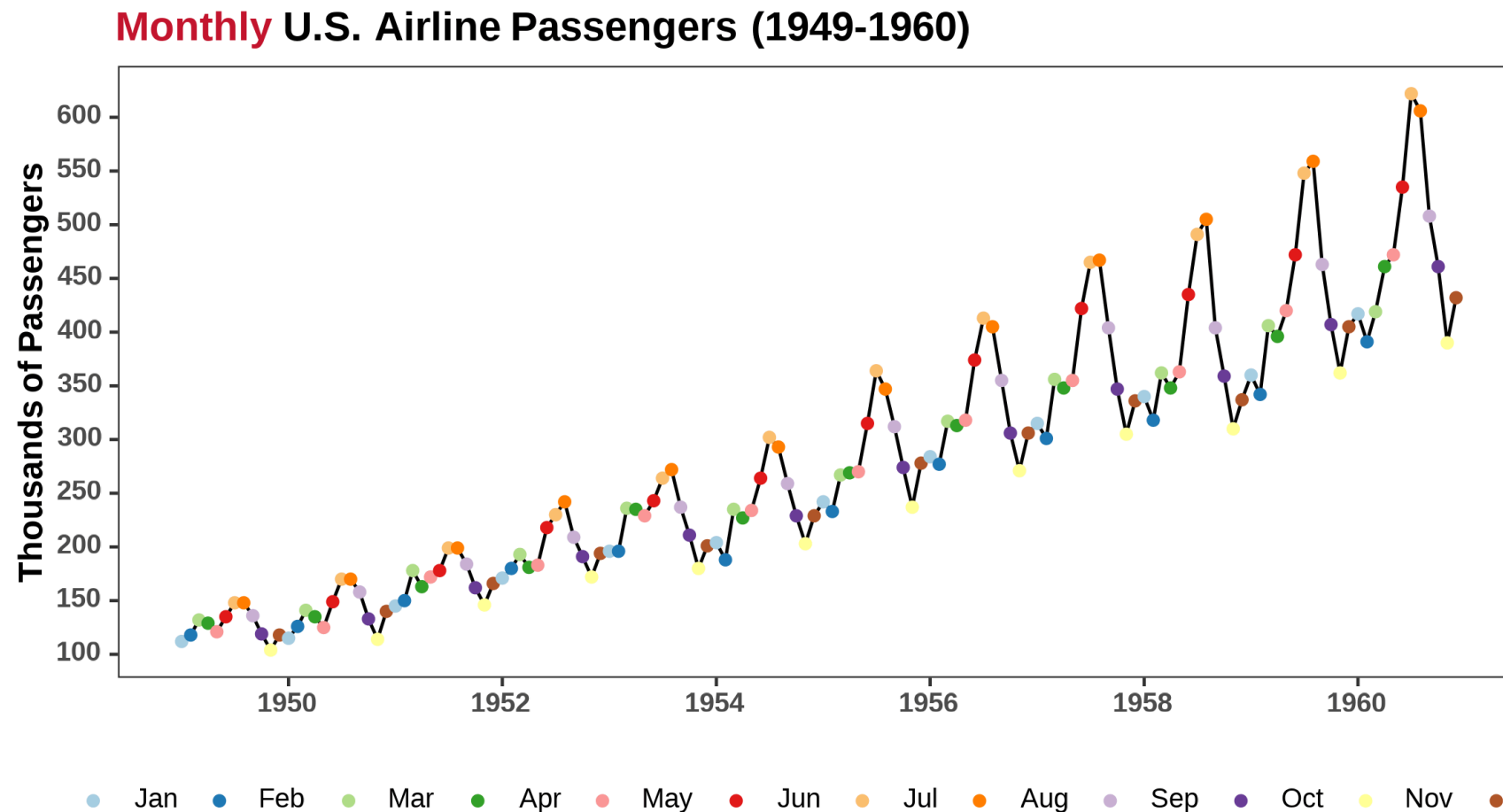
Stabilizing the Variance and the Mean: Log-Differencing

Log-differencing is a combination of **log transformation** and **differencing**.

Monthly U.S. Airline Passengers (1949-1960)



One Final Step: DeSeasonalizing the Time Series



Order Does Not Matter: Seasonal and First Differences

When both seasonal and first differences are applied ...

- it makes no difference which is done first---the result will be the same.

If $y'_t = y_t - y_{t-12}$ denotes seasonally differenced series, then twice-differenced series is

$$\begin{aligned} y_t^* &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-12}) - (y_{t-1} - y_{t-13}) \\ &= y_t - y_{t-1} - y_{t-12} + y_{t-13} . \end{aligned}$$

This is equivalent to the first-differenced series of the seasonally differenced series:

$$\begin{aligned} y_t^* &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-12}) - (y_{t-1} - y_{t-13}) \\ &= y_t - y_{t-1} - y_{t-12} + y_{t-13} . \end{aligned}$$

Why Do Seasonal Differencing First?

Hyndman & Athanasopoulos (2022) recommend applying seasonal differencing first because:

- If seasonal differencing alone makes the series stationary, we don't need further differencing.
- If we first apply regular differencing, the seasonal pattern may remain strong, meaning an extra seasonal differencing step will still be necessary.
- **Seasonal differencing is often more effective** at removing non-stationarity in data with **strong seasonality**.

Unit Root Tests

Unit Root Tests: ADF Test

- **Unit root tests** are used to determine whether a time series is **stationary** or **non-stationary**.
- The most common unit root test is the **Augmented Dickey-Fuller (ADF) test**.
- The null hypothesis of the ADF test is that the time series has a **unit root** (i.e., it is **non-stationary**).
- If the null hypothesis is rejected (e.g., if $p < 0.05$), the time series is considered **stationary**.
- The ADF test is available in Python through the `statsmodels` library, and can be accessed using the `adfuller()` function as follows:

```
from statsmodels.tsa.stattools import adfuller
```

Note: See [this link](#) for more information on the `adfuller()` function.

Unit Root Tests: KPSS Test

- The **Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test** is another unit root test used to determine whether a time series is **stationary** or **non-stationary**.
- The null hypothesis of the KPSS test is that the time series is **stationary** around a **fixed level**.
- If the null hypothesis is rejected (e.g., if $p < 0.05$), the time series is considered **non-stationary**.
- The KPSS test is available in Python through the `statsmodels` library, and can be accessed using the `kpss()` function as follows:

```
from statsmodels.tsa.stattools import kpss
```

Note: See [this link](#) for more information on the `kpss()` function.

CoreForecast's Short Cut Based on R's forecast Package

- In R, the `forecast` package provides a convenient function `ndiffs()` to determine the number of differences required to make a time series stationary.
- This function uses the **KPSS test** to determine the number of differences required to make the time series stationary.
- In `CoreForecast`, we can use the `num_diffs(x: ndarray, max_d: int = 1) -> int` function to determine the number of differences required to make a time series stationary. Similarly, we can use `num_seas_diffs(x: ndarray, season_length: int, max_d: int = 1) -> int` to determine the number of seasonal differences required to make a time series stationary.
- **Reference:** [CoreForecast Documentation](#).

Demo: Determining the Number of Differences Required

Let us examine the `airpassengers` dataset. Our goal is to make this dataset stationary by making the appropriate transformations.

Recap

Summary of Main Points

By now, you should be able to do the following:

- Define a **stationary time series**.
- Understand how plotting techniques can help identify non-stationarity.
- Understand how we can **stabilize a non-stationary time series to achieve stationarity** (differencing to stabilize the mean, and log transformation to stabilize the variance).
- Understand how to use **unit root tests** to test for stationarity.



Review and Clarification



- **Class Notes:** Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording:** The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Questions:** Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.